

Programming 2025/2026

Laboratory

Session 1 (April 14)

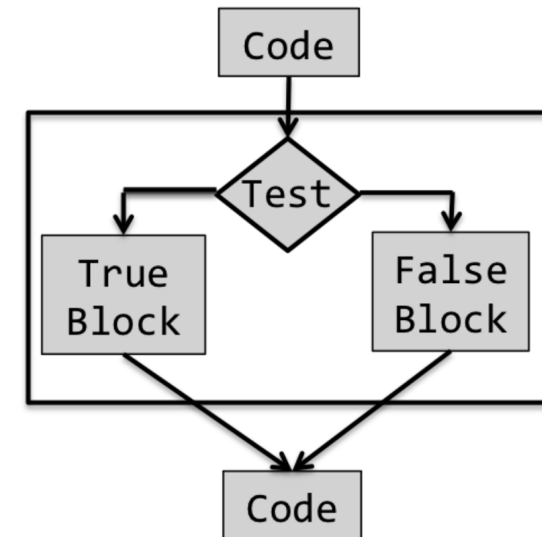
Branching and Input

First Cycle Degree/Bachelor in Genomics
Dept. Of Pharmacy and Biotechnology (FaBiT)
University of Bologna



Branching

```
if( condition ):  
    expression inside the if  
    ...  
    expression inside the if  
expression outside the if
```



- The condition is a Boolean expression which is evaluated to **True** or **False**
- If the condition is **True**, the program will **execute** the expressions inside the **if** statement (indentation!!!) and then the expressions outside the **if** statement
- If the condition is **False**, the program will **skip** the expressions inside the **if** statement and the execution passes to the expressions outside the **if** statement

Branching

```
age = 75
if (age >= 60):
    print("Over 60 reduced ticket")
```

Branching

- The condition can be a Boolean expression formed from a relational operator
 - $x == y$ (x equal to y)
 - $x != y$ (x different from y)
 - $x > y$ (x greater than y)
 - $x < y$ (x smaller than y)
 - $x >= y$ (x greater than or equal to y)
 - $x <= y$ (x smaller than or equal to y)
- The evaluation of the expressions above returns a Boolean
- **NOTE:** $x == 2$, $3 == x$ are valid expressions, but $4 = x$ is not a Boolean expression (and is not an assignment...try it)

Branching

- The condition can be an expression formed from a logical operator
 - `(a and b)` is `True` if both `a` and `b` are `True`
 - `(a or b)` is `True` if at least one of `a` and `b` is `True`
 - `(not a)` is `True` if `a` is `False`, and viceversa
- What are `a` and `b`?

Branching

```
if( condition ):
    expression inside the if
    ...
    expression inside the if
else:
    expression inside the else
    ...
    expression inside the else
expression outside the if-else
```

- The condition is a Boolean expression which is evaluated to **True** or **False**
- If the condition is **True**, the program will **execute** the expressions inside the **if** statement and then the expressions outside the **if-else** statement
- If the condition is **False**, the program will **execute** the expressions inside the **else** statement and then the execution passes to the expressions outside the **if-else** statement

Branching

```
age = 75
if (age >= 60):
    print("Over 60 reduced ticket")
else:
    print("Full ticket")
```


Branching

- The condition is a Boolean expression which is evaluated to **True** or **False**
- If the condition is **True**, the program will **execute** the expressions inside the **if** statement and then the expressions outside the **if-elif-else** statement
- If the condition is **False**, the program will evaluate the condition of the **first elif**
 - If the new condition is **True**, the program will **execute** the expressions inside the **first elif**, and then the expressions outside the **if-elif-else** statement
 - If the new condition is **False**, the program will evaluate the condition of the **second elif**
 - And so on...
- If all the conditions are **False**, the program will **execute** the expressions inside the **else** statement and after that the execution passes to the expressions outside the **if-elif-else** statement

```
if( condition ):
    expression inside the if
    ...
    expression inside the if
elif( condition ):
    expression inside the elif
    ...
    expression inside the elif
elif( condition ):
    expression inside the elif
    ...
    expression inside the elif
else:
    expression inside the else
    ...
    expression inside the else
expression outside the if-elif-else
```


Input

```
<var> = input("Some text describing the desired input")
```

E.g. `name = input("What is your name?")`

- The user will see the message `What is your name?` in the console
- The program will wait until the user inserts a value
- The variable `<var>` will contain a **string!**
 - Need to do type casting if another type (like `int`) is needed!

Input

```
answer = input("How old are you?")
if (answer < 18):
    print("You cannot get a driver license")
else:
    print("You can start preparing for you drive exams and practices!")
```

Does it work?

Input

```
answer = input("How old are you?")
if (answer < 18):
    print("You cannot get a driver license")
else:
    print("You can start preparing for you drive exams and practices!")
```

```
answer = int(input("How old are you?"))
if (answer < 18):
    print("You cannot get a driver license")
else:
    print("You can start preparing for you drive exams and practices!")
```



Exercise (2.1)

- Write a program that prints True or False when a given number is an even number or not

Exercise (2.2)

- Write a program that extends a previous example and prints:
 - “Free entrance” if the age is lower than 6
 - “Children reduced ticket” if the age is lower than 12
 - “Full ticket” if the age is between 12 and 60
 - “Over 60 reduced ticket” if the age is higher than 60
 - “Free entrance” if the age is higher than 70
- What happens when the age is 5?

Exercise (2.3)

- Given the following rent price table:

24 hours	2 days	3 days	4 days	Every extra day
50.00€	90.00€	120.00€	140.00€	30.00€

write a program that takes as input the number of days for which you want to rent a car and computes the final cost; the program works with float and if the number of days is lower than one it prints zero